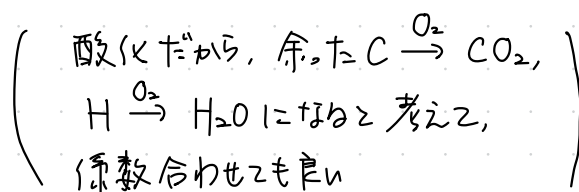
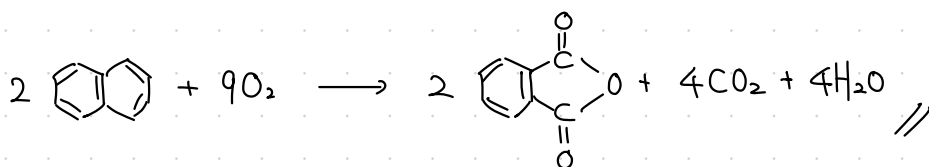
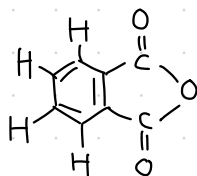
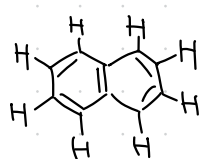
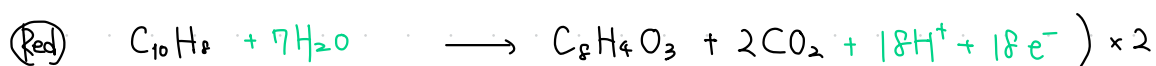
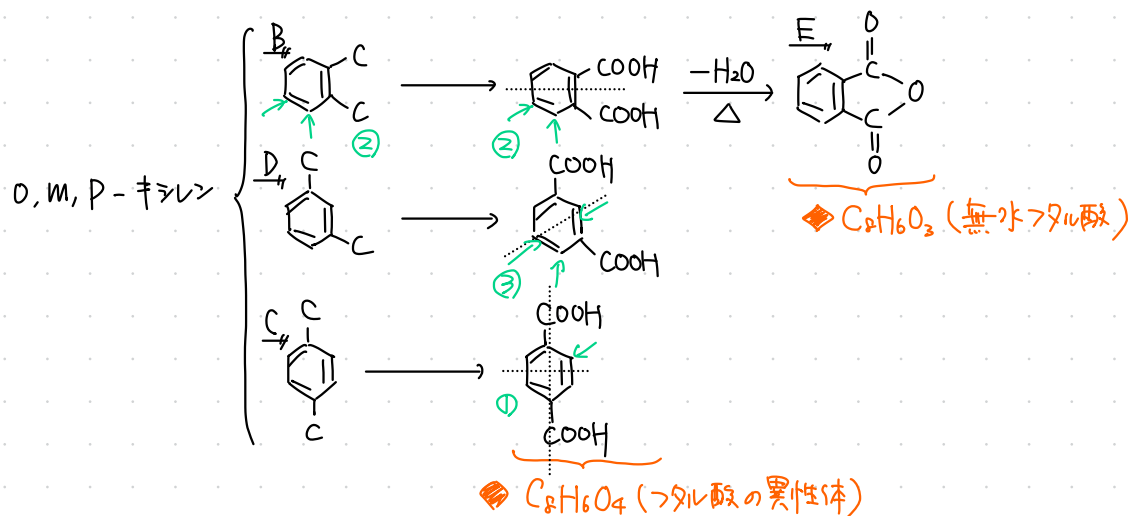
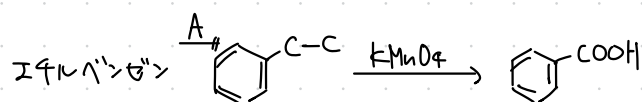


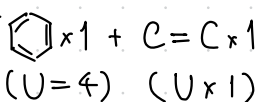
4-① C_8H_{10} $U = 4$ $\text{C}_6\text{H}_6 \times 1$



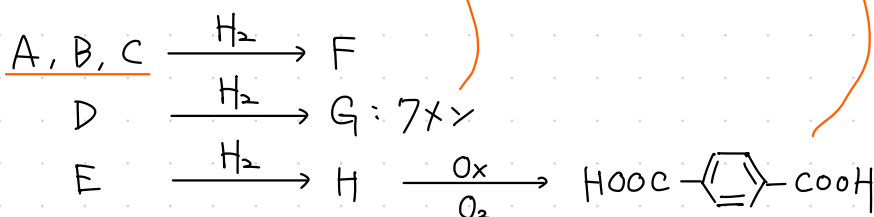
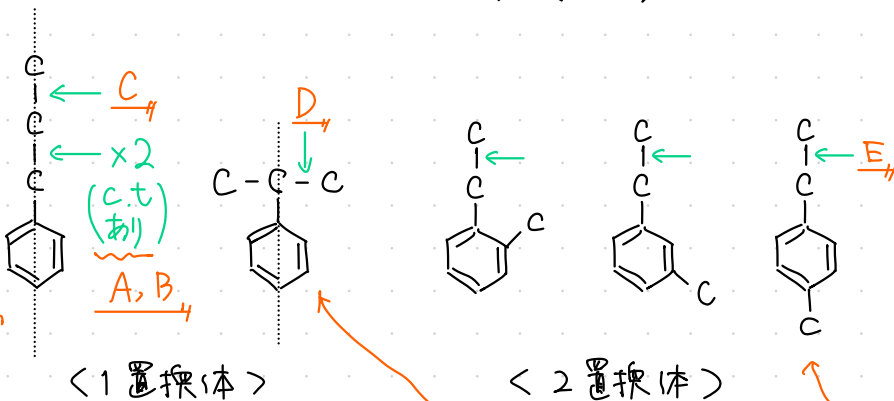
4-2

C_9H_{10}

$U = 5$

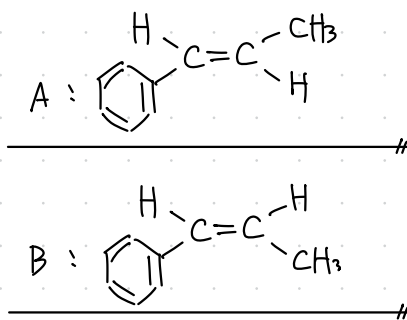


内2.



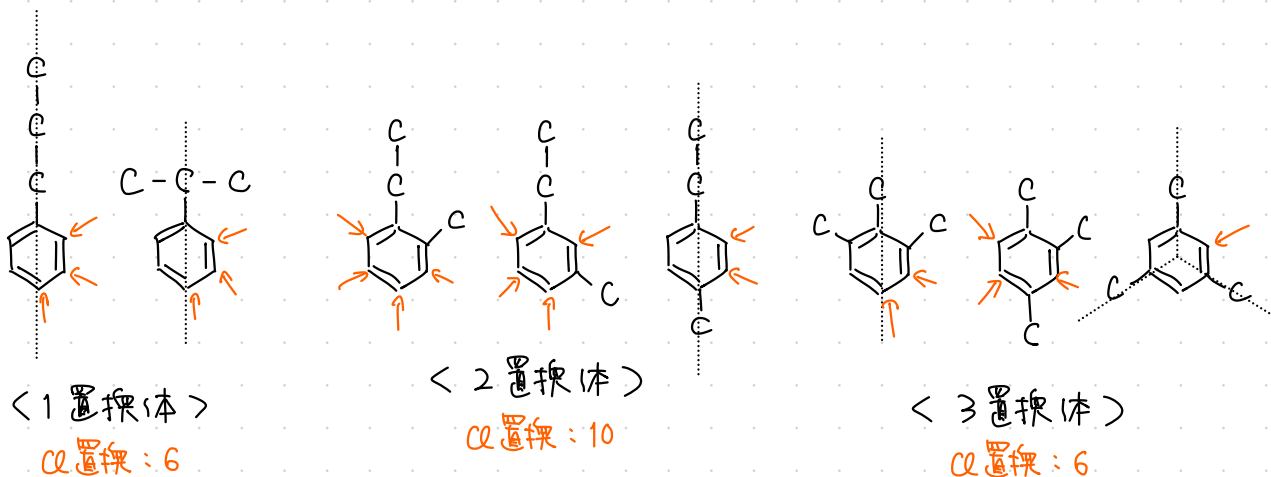
A, B: シス・トランス異性体

融点 $A > B$
 シス トランス



内1. 8

内3. 22



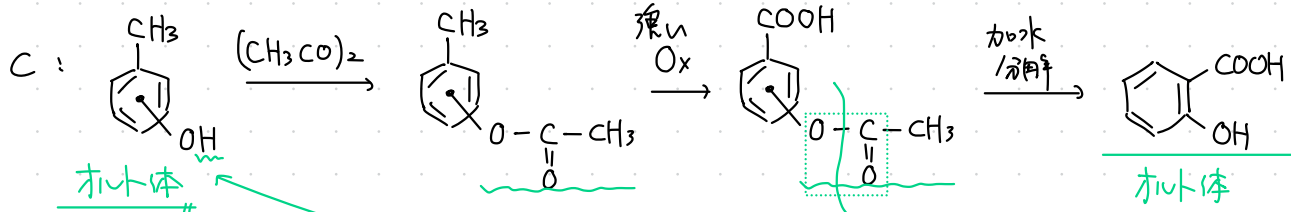
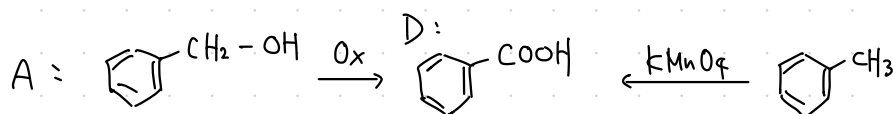
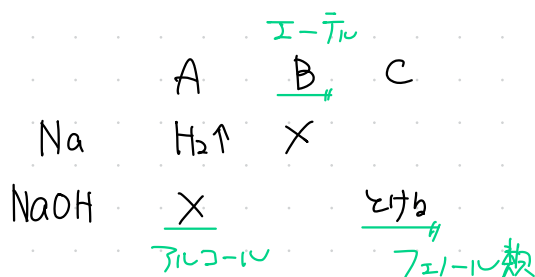
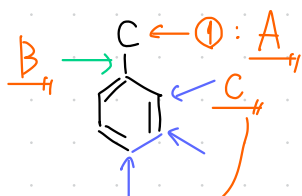
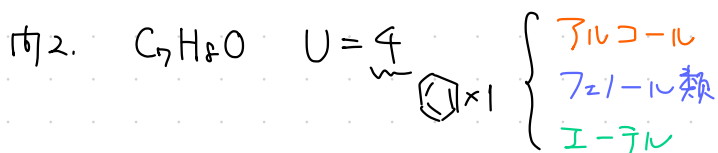
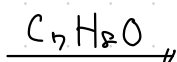
4-③

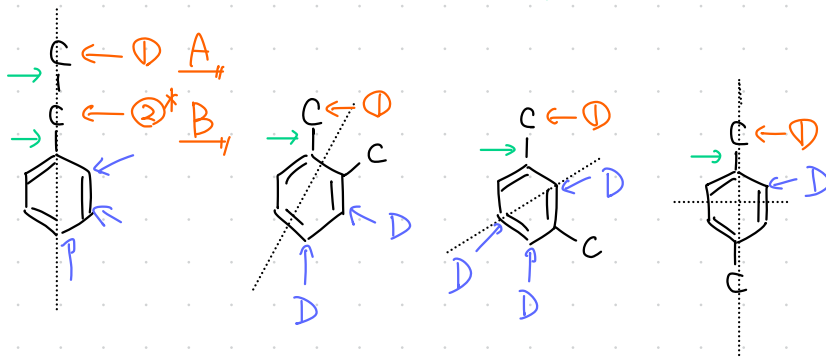
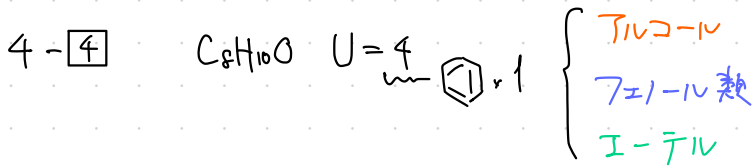


条件より $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

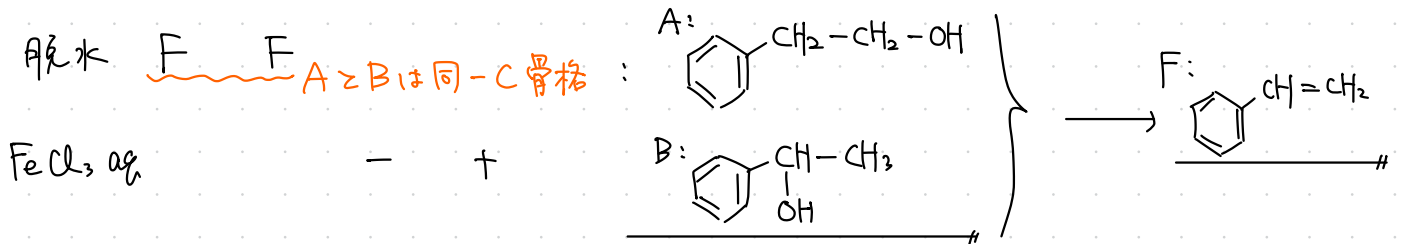
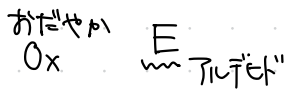
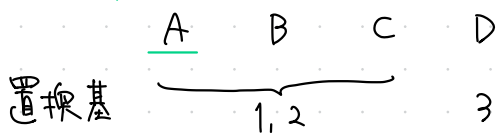
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

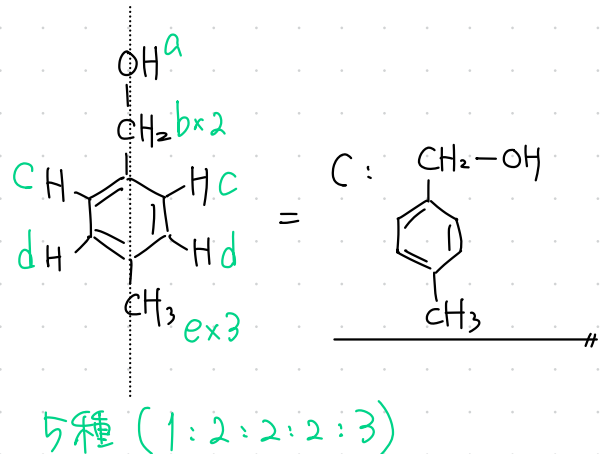
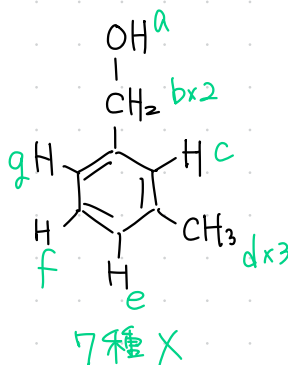
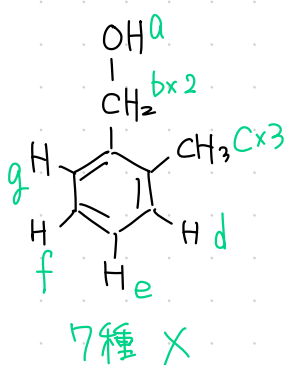




第1級
アルコール



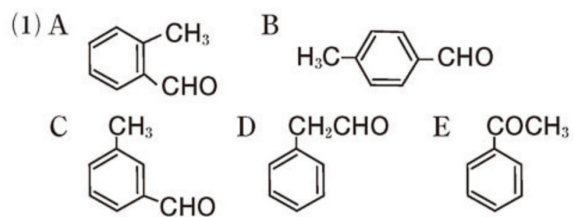
C : 1 or 2 置換体のアルコール



内2

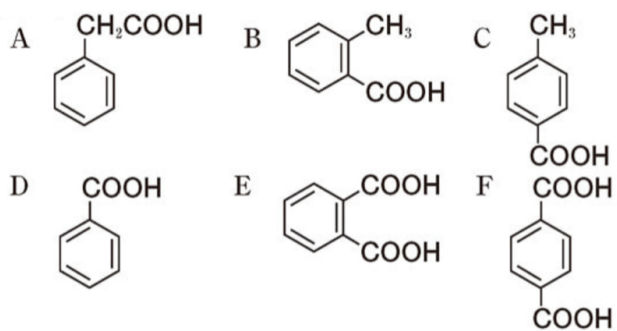
D: フェニル類 (3 置換体), $\underline{6}$

4 - 5

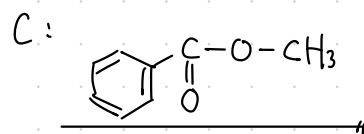
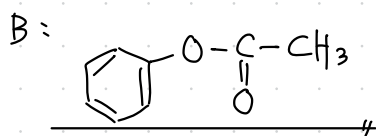
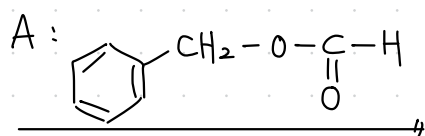
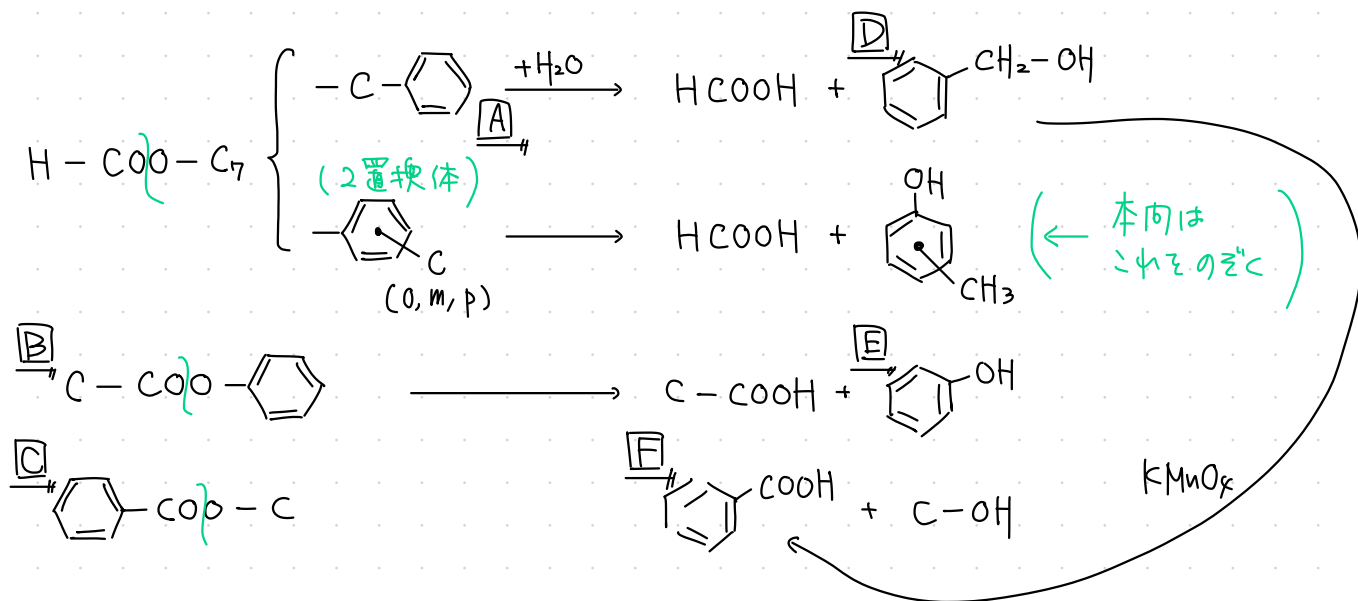
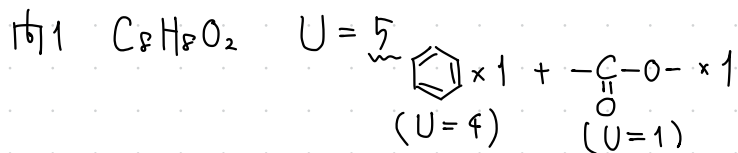


- (2) Gは*p*-置換体であるため、*o*-, *m*-置換体よりも分子の形が対称的である。したがって、結晶化しやすいため、融点は最も高くなる。

4 - 6

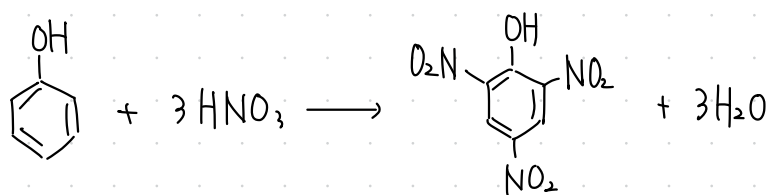


4-7



例2. E

例3



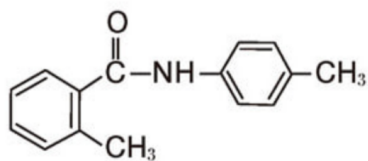
ヒョウリニ西ニ
 $(2, 4, 6 - \text{ト}) = \text{トロフエール}$

4 - 8

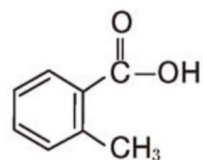
(1) $C_{15}H_{15}NO$

(2)

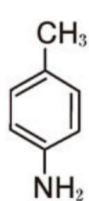
A



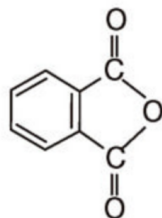
B



C

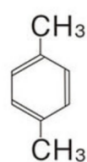


D

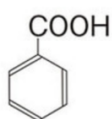


4 - 9

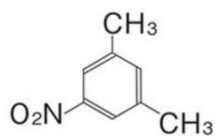
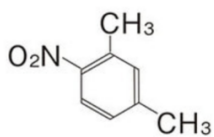
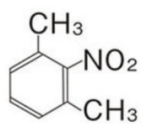
問 1 化合物 D :



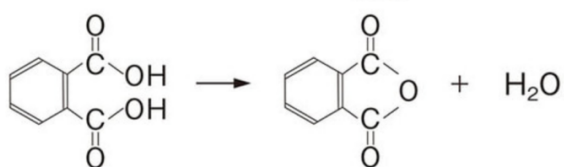
問 2 化合物 E :



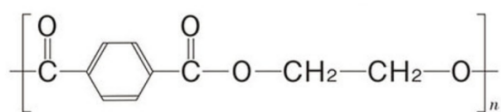
問 3



問 4



問 5

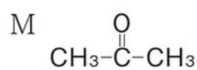
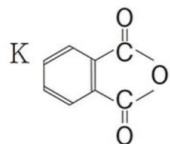
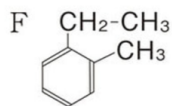
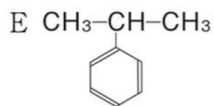
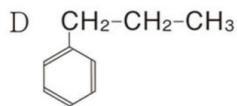
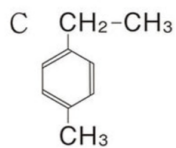
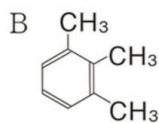
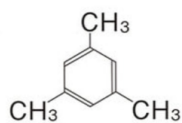


4 - 10

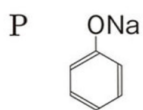
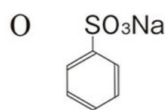
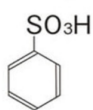
問1 (ア) エチレングリコール(1,2 エタンジオール) (イ) 酸素

(ウ) ヨウ素

問2 A



問3 N

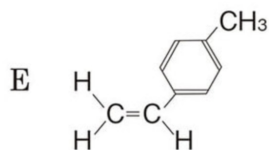
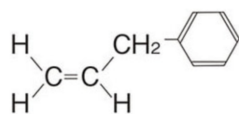


4 - 11

問1 2個

問2 クメン

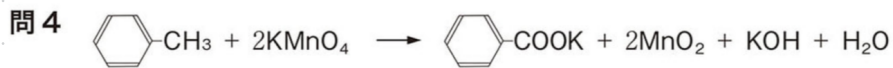
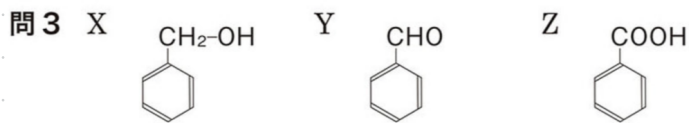
問3A



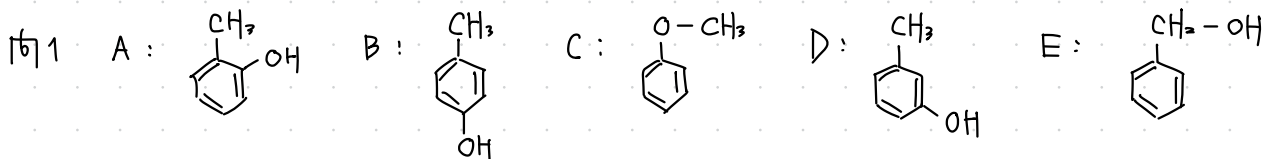
4-12

問1 気体…水素 官能基…ヒドロキシ基

問2 試薬… FeCl_3 色…紫



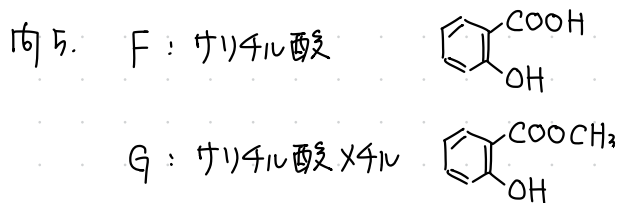
4-13



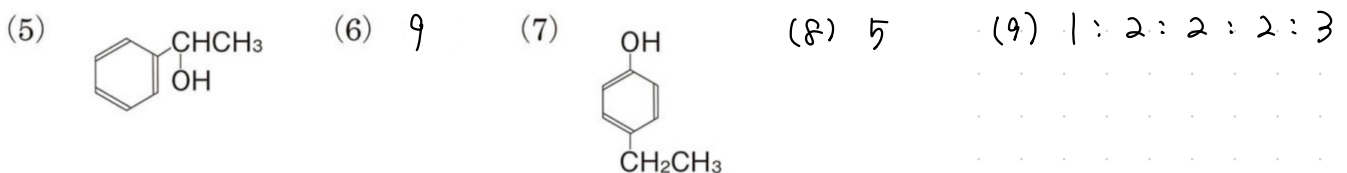
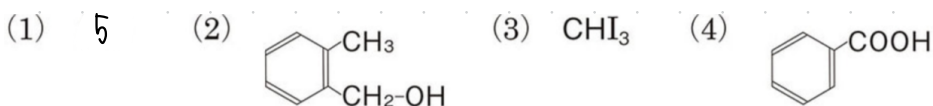
問2 NaOH aq を加えると A, B, D は 中和して均一な溶液になる (とける)。
(C, E は 2層に分離する) ↑ これは間違い。見ためや1においでわかることだから

問3 ヒドロキシ基 を持たないため、水素結合 ができず、分子間力が弱いから。

問4 金属 Na を加えたとき、C は H_2 を発生しない。

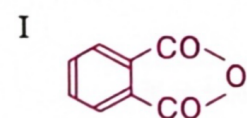
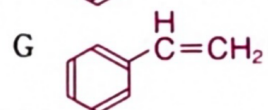
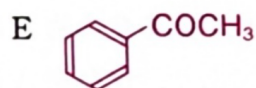
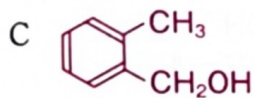
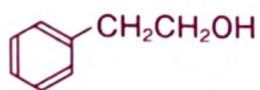


4-14

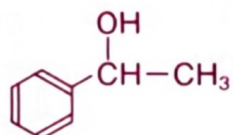


4 - 15

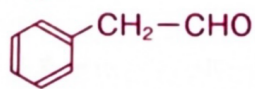
問1 A



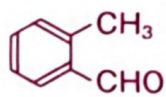
B



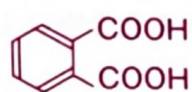
D



F



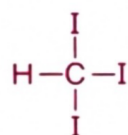
H



問2 B

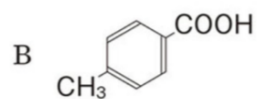
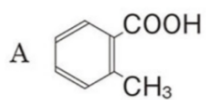
問3 ポリスチレン

問4 記号 B 構造式



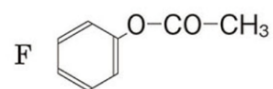
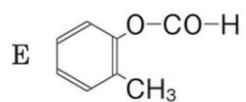
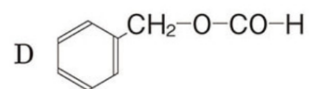
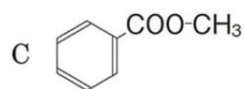
4 - 16

問1 (1)



(2) ナフタレン, *o*-キシレン

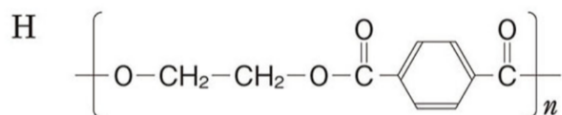
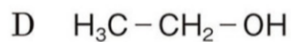
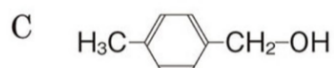
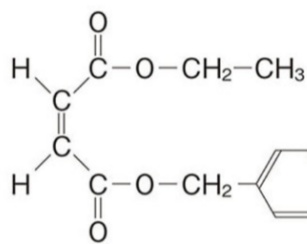
問2 (1)



4 - 17

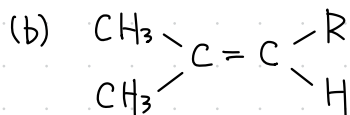
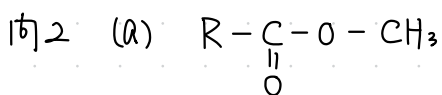
問1 $C_{14}H_{16}O_4$

問2 A

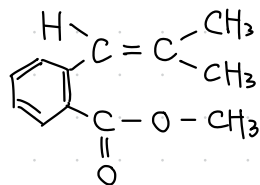


4 - 18

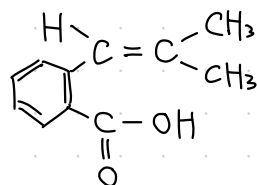
問1 $C_{12}H_{14}O_2$



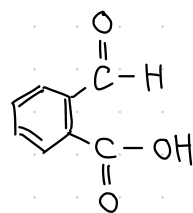
問3 A



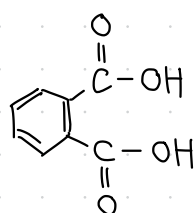
B



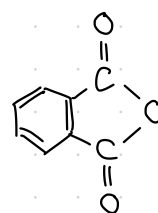
C



D

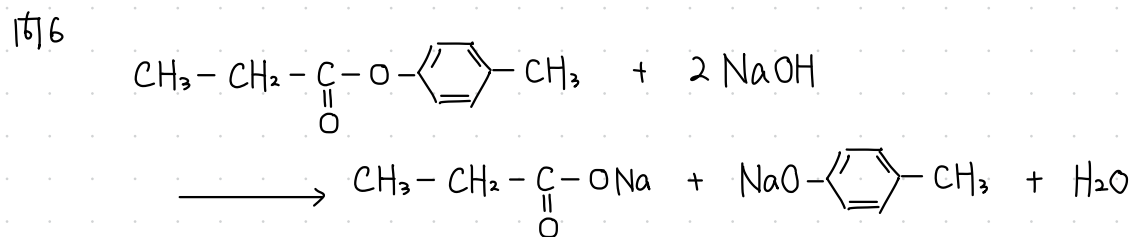
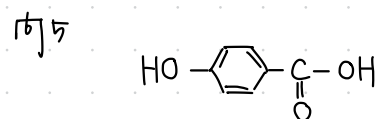
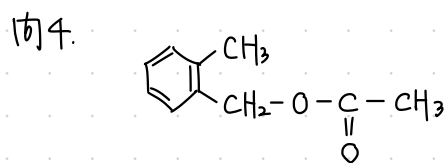
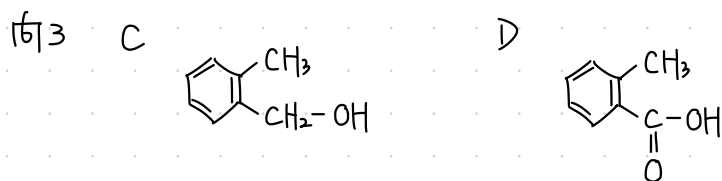
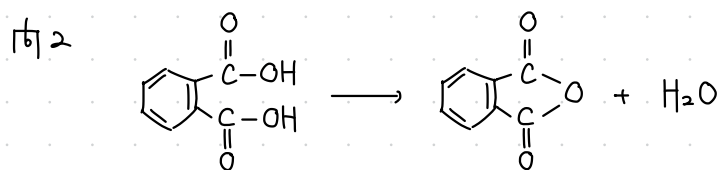


E



4 - 19

問1 あ C_5H_6O n 164 う $C_{10}H_{12}O_2$

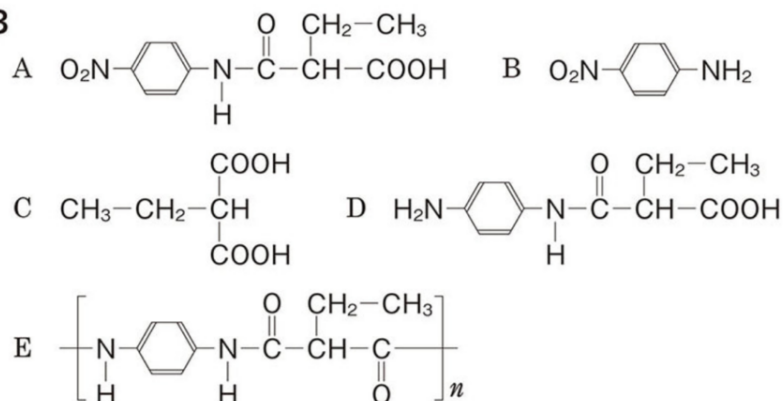


4- 20

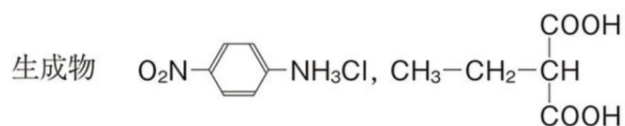
問1 C 原子…52.4 % H 原子…4.8 % N 原子…11.1 %

問2 $\text{RCOOH} + \text{NaHCO}_3 \rightarrow \text{RCOONa} + \text{CO}_2 + \text{H}_2\text{O}$

問3

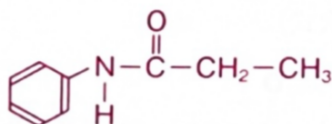


問4 塩酸を用いることで化合物 A のアミド結合が加水分解されてしまうため。

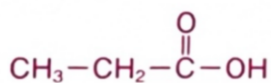


4- 21

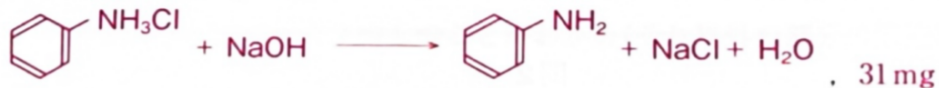
問1 A



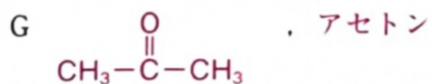
C



問2

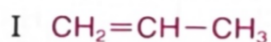
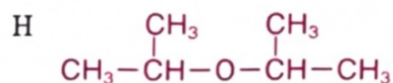


問3 F $\text{CH}_3\text{-CH}_2\text{-O-CH}_3$, エチルメチルエーテル



問4 F, E, D

問5



問1 $C_{19}H_{20}O_4$ 